

Zakriya Asif Paracha

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PROFILE

Software Engineer with 2 years of production experience building Ruby on Rails and PostgreSQL backend services and REST APIs serving millions of daily users. Strong in automated testing, production debugging, and data pipelines, with hands-on experience in AWS, Docker, CI/CD, and LLM application patterns such as RAG. Looking to build the application layer behind AI products and grow toward AI engineering.

WORK EXPERIENCE

Motive | L2 Software Engineer

Sep 2024 – Aug 2026

Ruby on Rails, Python, PostgreSQL, Redis, Apache Airflow, AWS (S3, IAM), Git, REST APIs

- Built and maintained Rails backend services and REST APIs on the Compliance team, serving millions of daily users.
- Operated a complex Apache Airflow pipeline (Python, SQL) transforming USA CSA panel data into user-facing compliance insights, including Airflow access management.
- Managed AWS S3 buckets and access permissions (IAM), applying least-privilege access for the team's data workflows.
- Wrote automated tests in a team effort that raised coverage from 60% to 80%+, reducing production regressions.
- Debugged and triaged critical production issues across the stack, reducing downtime and improving stability.
- Optimized PostgreSQL queries and Redis caching at scale; shipped through code review and sprint-based agile delivery.

BCUBE | Software Intern

Sep 2021 – Oct 2021

Python, PyQt, Device APIs

- Built a PyQt attendance system integrating a biometric device API, automating HR tracking and reporting.
- Implemented holiday and leave management with exportable attendance reports, replacing manual record-keeping for HR staff.

PROJECTS

CPSC Compliance Assistant (RAG on AWS)

2026 | [Live](#) | [GitHub](#)

Python, FastAPI, React, pgvector, LangChain, Amazon Bedrock, Docker, GitHub Actions

- Built the application around an LLM: a FastAPI streaming API, a Postgres/pgvector data layer, and a React UI that opens each citation at the exact source passage.
- Added hybrid vector + BM25 retrieval, citation checks, and a prompt-injection guard; an eval set lifted source recall from 10 to 16 of 17.
- Wrote 70 automated unit/integration tests run in CI, including against a real pgvector instance with the LLM calls faked.
- Deployed on AWS (EC2, S3, CloudFront, CloudFormation) via Docker and keyless GitHub OIDC CI/CD.

Warehouse Item Transportation via Multi-Hop Robotic Networks

Final Year Project, 2024

Python, Angular, MQTT, ESP8266

- Built a Python routing backend (ant colony optimization) and an Angular real-time dashboard; demoed on 3 robots.
- Designed a region-segmented architecture where robots hand items off at zone boundaries, removing most robot-to-robot collision-avoidance overhead.

Neural Network Inference on FPGA

2026

PyTorch, INT8 Quantization, Verilog, Basys 3

- Trained an MNIST classifier in PyTorch, quantized it to INT8 (94.57% accuracy), and ran inference in Verilog in 0.51 ms.

SKILLS

Backend:	Ruby on Rails, Python, FastAPI, Flask, REST APIs, Apache Airflow
Databases:	PostgreSQL, pgvector, Redis, SQL
AI / LLM:	RAG, embeddings, hybrid search, LangChain, Amazon Bedrock, LLM evaluation, prompt-injection guardrails
Cloud & DevOps:	AWS (S3, IAM, EC2, CloudFront, CloudFormation, Bedrock), Docker, GitHub Actions CI/CD, Linux, Git
Testing:	pytest, RSpec, unit and integration testing
Frontend:	React, TypeScript, JavaScript, HTML, CSS, Angular
Also:	Data Structures & Algorithms, API Design, System Design; Go, C/C++, Verilog, PyTorch

EDUCATION

CEME, National University of Sciences and Technology (NUST), Islamabad

BE Computer Engineering

June 2024

CGPA: 3.66/4.0

COURSES & CERTIFICATIONS

AWS Cloud Solutions Architect (*in progress; Cloud Technical Essentials completed*)

AWS – Coursera Specialization

Google Cybersecurity

Google – Coursera Specialization

Full list of 30+ courses: zakriyaparacha.com/courses.html